

EXT 1: Vectors (Ext1), V1 Introduction to Vectors (Y12)
Further Operations With Vectors (Ext1)

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Exam Equivalent Time: 24 minutes (based on HSC allocation of 1.5 minutes approx. per mark)

Questions

1. Vectors, EXT1 V1 2015 SPEC2 15 MC

The projection of the force $\underline{F} = a\underline{i} + b\underline{j}$, where a and b are non-zero real constants, in the direction of the vector $\underline{w} = \underline{i} + \underline{j}$, is

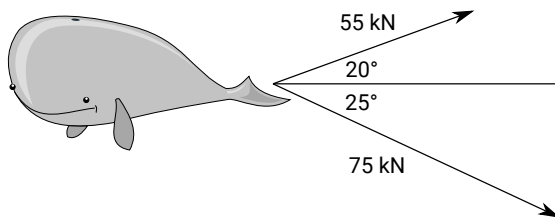
- A. $\left(\frac{a+b}{2}\right)\underline{w}$
- B. $\frac{\underline{F}}{a+b}$
- C. $\left(\frac{a+b}{a^2+b^2}\right)\underline{F}$
- D. $\left(\frac{a+b}{\sqrt{2}}\right)\underline{w}$

2. Vectors, EXT1 V1 SM-Bank 12

A beached whale is being rescued by two jet-skis with the forces measured in kilonewtons (kN).

The angle of the forces are measured against an east-west axis line.

What is the resultant force of the two jet-skis in magnitude and direction? (3 marks)

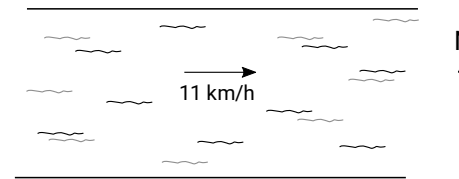


3. Vectors, EXT1 V1 EQ-Bank 3

A force described by the vector $\underline{F} = \begin{pmatrix} 3 \\ 6 \end{pmatrix}$ newtons is applied to a line $\underline{l}$ which is parallel to the vector $\begin{pmatrix} 4 \\ 3 \end{pmatrix}$.

- i. Find the component of the force $\underline{F}$ in the direction of $\underline{l}$. (2 marks)
- ii. What is the component of the force $\underline{F}$ in the direction perpendicular to the line? (1 mark)

4. Vectors, EXT1 V1 SM-Bank 10



A man attempts to swim north across a river at a speed of 4.5 km/h.

If the river's current is moving at 11 km/h due east, find the bearing (nearest degree) and speed (to 1 decimal place) that the man will actually be swimming. (3 marks)

5. Vectors, EXT1 V1 2021 HSC 14a

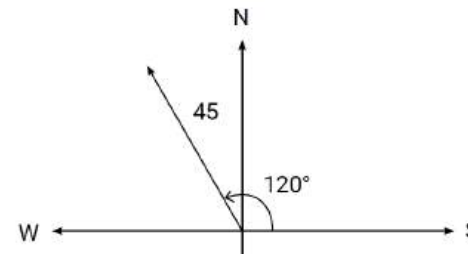
A plane needs to travel to a destination that is on a bearing of 063° . The engine is set to fly at a constant 175 km/h. However, there is a wind from the south with a constant speed of 42 km/h.

On what constant bearing, to the nearest degree, should the direction of the plane be set in order to reach the destination? (3 marks)

6. Vectors, EXT1 V1 SM-Bank 11

A drone is set to fly west at 38 km/h.

A cross wind diverts its path so that it travels with a speed of 45 km/h in the direction shown below.



Calculate the speed, to one decimal place, and the bearing of the cross wind, to the nearest degree. (3 marks)

Worked Solutions

1. Vectors, EXT1 V1 2015 SPEC2 15 MC

$$\hat{w} = \frac{\tilde{w}}{\sqrt{1+1}}$$

$$= \frac{\tilde{i} + \tilde{j}}{\sqrt{2}}$$

$$\tilde{F} \cdot \hat{w} = \frac{a+b}{\sqrt{2}}$$

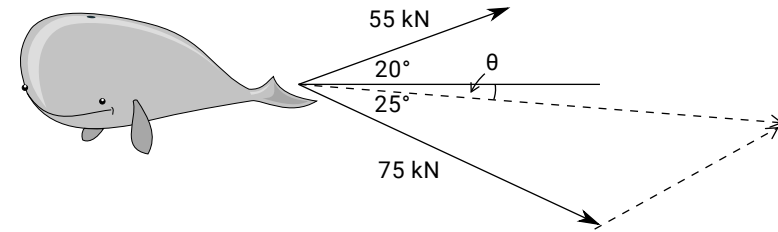
$$\begin{aligned} (\tilde{F} \cdot \hat{w})\hat{w} &= \left(\frac{a+b}{\sqrt{2}}\right) \frac{\tilde{w}}{\sqrt{2}} \\ &= \left(\frac{a+b}{2}\right) \tilde{w} \end{aligned}$$

$\Rightarrow A$

♦ Mean mark 49%.

Worked Solutions

2. Vectors, EXT1 V1 SM-Bank 12



Split each force into component forces:

$$\underline{u} = (55 \cos 20, 55 \sin 20)$$

$$\underline{v} = (75 \cos 25, -75 \sin 25)$$

$$\underline{F}_{\text{net}} = \underline{u} + \underline{v} = \begin{pmatrix} 55 \cos 20 + 75 \cos 25 \\ 55 \sin 20 - 75 \sin 25 \end{pmatrix}$$

$$\begin{aligned} |\underline{F}_{\text{net}}| &= \sqrt{(55 \cos 20 + 75 \cos 25)^2 + (55 \sin 20 - 75 \sin 25)^2} \\ &= 120.347... \end{aligned}$$

$$\tan \theta = \frac{|55 \sin 20 - 75 \sin 25|}{55 \cos 20 + 75 \cos 25}$$

$$= 0.10768$$

$$\theta = 6^\circ$$

$\therefore$ The resultant force is 120 kN on a bearing of 096° .

3. Vectors, EXT1 V1 EQ-Bank 3

i. $\underline{\underline{F}} = \begin{pmatrix} 3 \\ 6 \end{pmatrix}, \underline{\underline{v}} = \begin{pmatrix} 4 \\ 3 \end{pmatrix}$

$$\hat{\underline{\underline{v}}} = \frac{\underline{\underline{v}}}{|\underline{\underline{v}}|} = \frac{\underline{\underline{v}}}{\sqrt{4^2 + 3^2}} = \frac{1}{5}\underline{\underline{v}} = \begin{pmatrix} 0.8 \\ 0.6 \end{pmatrix}$$

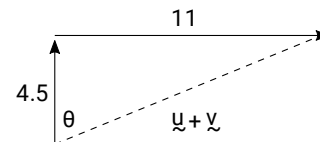
$$\begin{aligned}\underline{\underline{F}} \cdot \hat{\underline{\underline{v}}} &= \begin{pmatrix} 3 \\ 6 \end{pmatrix} \cdot \begin{pmatrix} 0.8 \\ 0.6 \end{pmatrix} \\ &= 3 \times 0.8 + 6 \times 0.6 \\ &= 6\end{aligned}$$

$$\begin{aligned}\text{proj}_{\underline{\underline{v}}} \underline{\underline{F}} &= (\underline{\underline{F}} \cdot \hat{\underline{\underline{v}}}) \hat{\underline{\underline{v}}} \\ &= 6 \begin{pmatrix} 0.8 \\ 0.6 \end{pmatrix} \\ &= \begin{pmatrix} 4.8 \\ 3.6 \end{pmatrix}\end{aligned}$$

ii. Component of $\underline{\underline{F}} \perp l$

$$\begin{aligned}&= \begin{pmatrix} 3 \\ 6 \end{pmatrix} - \begin{pmatrix} 4.8 \\ 3.6 \end{pmatrix} \\ &= \begin{pmatrix} -1.8 \\ 2.4 \end{pmatrix}\end{aligned}$$

4. Vectors, EXT1 V1 SM-Bank 10



$$\underline{\underline{u}} = (0, 4.5)$$

$$\underline{\underline{v}} = (11, 0)$$

$$\underline{\underline{u}} + \underline{\underline{v}} = \begin{pmatrix} 0 \\ 4.5 \end{pmatrix} + \begin{pmatrix} 11 \\ 0 \end{pmatrix} = \begin{pmatrix} 11 \\ 4.5 \end{pmatrix}$$

$$\tan \theta = \frac{11}{4.5}$$

$$\theta = \tan^{-1} \left(\frac{11}{4.5} \right)$$

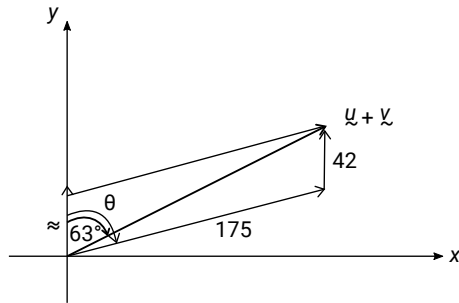
$$\theta = 67.75$$

$$\theta \approx 068^\circ$$

$$\begin{aligned}|\underline{\underline{u}} + \underline{\underline{v}}| &= \sqrt{11^2 + 4.5^2} \\ &= 11.88. . \\ &= 11.9 \text{ km/h (1 d.p.)}\end{aligned}$$

$\therefore$ Swimmer's speed is 11.9 km/h on a bearing of 068°

5. Vectors, EXT1 V1 2021 HSC 14a



$$\underline{u} = (175 \sin \theta, 175 \cos \theta)$$

$$\underline{v} = (0, 42)$$

$$\underline{u} + \underline{v} = (175 \sin \theta, 175 \cos \theta + 42)$$

$$\therefore \tan 63^\circ = \frac{175 \sin \theta}{175 \cos \theta + 42}$$

$$(175 \cos \theta + 42) \tan 63^\circ = 175 \sin \theta$$

$$175 \cos \theta \tan 63^\circ + 42 \tan 63^\circ = 175 \sin \theta$$

$$175 \sin \theta - 175 \cos \theta \tan 63^\circ = 42 \tan 63^\circ$$

$$\sin \theta - \cos \theta \frac{\sin 63^\circ}{\cos 63^\circ} = \frac{6}{25} \frac{\sin 63^\circ}{\cos 63^\circ}$$

$$\sin \theta \cos 63^\circ - \cos \theta \sin 63^\circ = \frac{6}{25} \sin 63^\circ$$

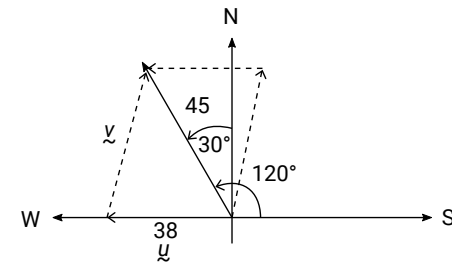
$$\sin(\theta - 63^\circ) = \frac{6}{25} \sin 63^\circ$$

$$\theta - 63^\circ = \sin^{-1}\left(\frac{6}{25} \sin 63^\circ\right)$$

$$\theta - 63^\circ = 12^\circ$$

$$\therefore \theta \approx 75^\circ$$

6. Vectors, EXT1 V1 SM-Bank 11



$$\underline{u} + \underline{v} = (-45 \sin 30^\circ, 45 \cos 30^\circ)$$

$$= \left(-22.5, \frac{45\sqrt{3}}{2}\right)$$

$$\underline{u} = (-38, 0), \quad \underline{v} = (x, y)$$

$$\begin{pmatrix} -38 \\ 0 \end{pmatrix} + \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -22.5 \\ \frac{45\sqrt{3}}{2} \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 15.5 \\ \frac{45\sqrt{3}}{2} \end{pmatrix}$$

$$|\underline{v}| = \sqrt{15.5^2 + \frac{45^2 \times 3}{4}}$$

$$= 41.94...$$

$$= 41.9 \text{ km/h}$$

$$\tan \theta = \frac{\frac{45\sqrt{3}}{2}}{15.5} = 2.514$$

$$\theta = 68.3^\circ$$

$\therefore$ The crosswind has a speed of 41.9 km/h on a bearing of 022° .